

The value of $\lambda(4)$

A machine-derived proof that $\lambda(4) = -L(1,2,3,4)$, completing the program of Mercer's Section 5.

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Status

This proof was derived and verified by a certified-arithmetic engine and has **not** been peer-reviewed. Every constant in this document is interpolated from the machine record at build time; the mathematical prose (lemma proofs, cone bijections, the proof template) is authored and should be reviewed like any mathematics. An independent audit walk (direct trigonometric summation, no code shared with the engine) covered every gcd-reduced 4-set with max element ≤ 30 — 25819 sets, every one reached by an explicit clause of this proof, 423 finite-clause sets re-certified fresh, and a full sweep found zero refuters. Reproduction: `node tools/run-lambda4-campaign.js` (the whole campaign, ~30 min), `node instruments/lambda4/battery.js` (calibration + red controls, ~1 s), `node tools/audit-lambda4.js` (the independent walk).

1. Statement

For a set $A = \{a_1 < \dots < a_n\}$ of positive integers write $f_A(t) = \sum \cos(a_i t)$ and $L(A) = \min_t f_A(t) < 0$. Following Mercer [M], $-\lambda(n) = \sup_A L(A)$ over all n -element sets; $\lambda(n)$ is the shallowest possible dip. Mercer proved $\lambda(2) = 9/8$ and $\lambda(3) = (17 + 7\sqrt{7})/27$, and conjectured:

Theorem. $\lambda(4) = -L(1,2,3,4)$. Numerically $\lambda(4) = 1.5195578816428\dots$; exactly, $\lambda(4)$ is the root of

$$512 y^3 - 1227 y^2 + 600 y + 125 = 0$$

in the certified enclosure $[-855435038691829/562949953421312, -6843480309534631/4503599627370496]$ negated. Equality holds for $\{1,2,3,4\}$ and its dilations and, by strictness of every closure below, for no other set.

The extremal value is an algebraic number of degree 3: with $c = \cos t$, $f_{\{1,2,3,4\}} = 8c^4 + 4c^3 - 6c^2 - 2c$, and the cubic above is the exact resultant of this quartic minus y against its derivative, restricted to the enclosure. Both the enclosure (BigInt Sturm isolation + interval Newton) and the sign change of the cubic across it (exact rational evaluation) are re-proved mechanically at every build of this document's companion page.

Since gcd-scaling leaves $L(A)$ unchanged ($f_{\{kA\}}(t) = f_A(kt)$), it suffices to prove $\min f_A \leq -L(1,2,3,4)$ for every $A = \{a < b < c < d\}$ with $\gcd(A) = 1$ — the domain N'_4 .

2. The toolkit

2.1 Averages over equispaced sets

For $m \geq 1$ and an anchor x_i , let $S = S(m, x_i) = \{ t : m t = x_i \pmod{2\pi} \}$, a set of m equally spaced points. For the average $\langle g \rangle_S = (1/m) \sum_{t \in S} g(t)$:

Lemma A (the average of a cosine). $\langle \cos(k t) \rangle_S = \cos(t' x_i)$ if $k = t' m$ for a nonnegative integer t' , and 0 otherwise.

Proof. With $t_j = (x_i + 2\pi j)/m$, $\sum_j \cos(k t_j) = \operatorname{Re} [e^{i k x_i / m} \sum_j e^{i 2\pi j k / m}]$, and the inner geometric sum is m when $m \mid k$ and 0 otherwise; when $k = t' m$ the phase is $e^{i t' x_i}$. $\square$

Products expand by $\cos A \cos B = (\cos(A+B) + \cos(A-B))/2$, so the inner product $\langle w, g \rangle_S$ of two cosine polynomials is a finite rational combination of Lemma-A averages. The anchors used are x_i in $\{0, \pi/6, \pi/3, \pi/2, 2\pi/3, \pi, 4\pi/3\}$; at every multiple this proof evaluates, $\cos(t' x_i)$ is rational (at $\pi/6$ only even multiples and $t' = 3, 9 \pmod{12}$ occur, all rational: $0, \pm 1/2, \pm 1$).

2.2 The piecewise structure over a cone

Every family below is presented by a **cone**: members are linear forms with nonnegative integer coefficients over free parameters $x_1, \dots, x_k \geq 1$. Two elementary principles make Lemma A computable uniformly over a cone:

- **(Positivity.)** A form with nonnegative coefficients summing to ≥ 1 is ≥ 1 at every point of the cone (its minimum is at $x = (1, \dots, 1)$).
- **(Bounded multiples.)** If $Bm - k$ has nonnegative coefficients summing to ≥ 1 , then $k < Bm$ on the cone, so $k = t' m$ is possible only for $t' < B$ — finitely many candidate linear conditions, each kept only if neither it nor its negation is certified positive.

Consequently $\langle w, g \rangle_S$, as a function on the cone, equals an exact rational **base value** plus, for each of finitely many linear **collision conditions**, an exact rational **delta** added when that condition holds. Additivity over simultaneously active conditions is exact, not approximate: each Lemma-A average depends only on its own divisibility condition. Forms of unknown sign are handled through evenness of cosine: $|k| = t' m$ with the vanishing case $t' = 0$ (contributing $\cos 0 = 1$) included as a genuine condition.

2.3 The dip extraction

Lemma B (Mercer's Lemma 3.4, with the hypothesis made explicit). If $w \geq 0$ on the circle, $\langle w, g \rangle_S \leq 0$, and $\langle w, 1 \rangle_S > 0$, then $g(t) \leq 0$ for some t in S .

Proof. If $g > 0$ everywhere on S , then $\sum w(t) g(t) \geq 0$ with equality only if $w g$ vanishes identically on S , forcing $w = 0$ on S and $\langle w, 1 \rangle_S = 0$. $\square$

The engine verifies $\langle w, 1 \rangle_S > 0$ on **every** branch it closes: the worst case over all collision subsets of the positivity expression (conservative, since inconsistent subsets only help). All

weights are drawn from the nonnegative cone generated by constants, $(1 - \cos k t)$, $(1 + \cos k t)$ and their squares.

2.4 The two-set estimate

Lemma C (Mercer's Lemma 3.1). *Equispaced sets $S(m_1, x_{i_1})$, $S(m_2, x_{i_2})$ with $g = \gcd(m_1, m_2)$ contain points t_1, t_2 with $|t_1 - t_2| \leq \pi g / (m_1 m_2)$.*

Whenever Lemma C is invoked, $g = 1$ is **certified**, never assumed: every member of the (gcd-reduced) set is exhibited as an integer combination of the two orders, so a common divisor of the orders would divide $\gcd(A) = 1$.

Evaluating at the $S(m_1, x_{i_1})$ -point t_1 provided by Lemma C: $m_1 t_1 = x_{i_1}$ exactly, and

$$m_2 t_1 - x_{i_2} \leq \pi / m_1. \text{ A member } k = u m_1 + v m_2 \text{ then satisfies}$$

$k t_1 = u x_{i_1} + v x_{i_2} + v \delta$ with $|v \delta| \leq |v| \pi / N$, N the value of m_1 . Each member is finished by one of:

- **exact** ($v = 0$): $\cos(k t_1) = \cos(u x_{i_1})$, a rational from the anchor table;
- **Lemma D** (target π): $\cos(\pi + e) \leq -1 + e^2/2$ for all e (since $\cos e \geq 1 - e^2/2$);
- **Lemma E** (target $2\pi/3$ or $4\pi/3$; Mercer's Lemma 3.3): $\cos(2\pi/3 \pm e) \leq -1/2 + (3/\pi) e$ for $0 \leq e \leq \pi/6$. *Proof.* On $[\pi/2, 2\pi/3]$, $\cos'' = -\cos \geq 0$, so $\cos$ is convex and lies below the chord through $(\pi/2, 0)$ and $(2\pi/3, -1/2)$, whose equation is $-(3/\pi)(t - \pi/2)$; at $t = 2\pi/3 - e$ this is $-1/2 + (3/\pi)e$. For the right side $\cos(2\pi/3 + e) \leq -1/2 \leq -1/2 + (3/\pi)e$. The $\pi/6$ range restriction keeps the left side inside $[\pi/2, 2\pi/3]$. $\square$ The validity range is carried as a floor on every derived threshold.
- **Lemma F** (target $\pi/2$ or $3\pi/2$, new here): $\cos(\pi/2 \pm e) \leq e$ and $\cos(3\pi/2 \pm e) \leq e$, for all e , since $|\cos(\pi/2 + x)| = |\sin x| \leq |x|$.

Summing gives a certified bound $f_A(t_1) \leq A_q + B_q \pi^2/N^2 + C_q/N + D_q \pi/N$ with exact rationals $A_q, B_q, C_q, D_q \geq 0$ (except A_q), decreasing in N . The **derived threshold** N_0 is the least integer at or above every validity floor whose bound clears the target $L(1,2,3,4)$ (compared through a directed rational enclosure of π — floating point never enters a decision). Below N_0 the family is a finite list, and every member of it is decided by the certified minimum instrument (Chebyshev reduction, BigInt Sturm isolation, interval Newton — calibrated by re-deriving Mercer's $\lambda(2)$ and $\lambda(3)$ at every test run, thresholds $a \geq 3$, $b \geq 3$, $b \geq 33$ re-derived, 324 calibration sets decided).

3. The generic case

On $S(d, \pi)$ take the weight $w = (1 - \cos a t) + (1 - \cos b t) + 2(1 - \cos c t)^2$ and the target $g = 3/5 + \cos a t + \cos b t + \cos c t$. The engine computes $\langle w, g \rangle_S = 0$ and $\langle w, 1 \rangle_S = 5$ in base, so by Lemma B, $\cos a t + \cos b t + \cos c t \leq -3/5$ somewhere on S ; adding $\cos(d t) = -1$ there gives $\min f_A \leq -8/5 < L(1,2,3,4)$. The computation is valid except on the collision conditions, which the

engine derives (they are OUTPUT — the fourteen match Mercer's hand-written list exactly) together with each one's exact delta:

condition	delta	consequence	example
2a = d	1/2	must be handled	{3,4,5,6}
2b = d	1/2	must be handled	{1,2,3,4}
2c = a+d	-1/2	closes itself	{1,2,3,5}
2c = b+d	-1/2	closes itself	{1,2,3,4}
2c = d	7/5	must be handled	{1,2,3,6}
3c = 2d	1/2	must be handled	{1,2,4,6}
3c = d	-1/2	closes itself	{1,2,3,9}
a+2c = 2d	1/2	must be handled	{2,3,4,5}
a+2c = d	-1/2	closes itself	{1,2,3,7}
a+b = d	1	must be handled	{2,3,4,5}
a+c = d	5/2	must be handled	{1,2,3,4}
b+2c = 2d	1/2	must be handled	{1,2,3,4}
b+2c = d	-1/2	closes itself	{1,2,3,8}
b+c = d	5/2	must be handled	{1,2,3,5}

Five of the fourteen carry strictly negative delta: any set activating only those still has $\langle w, g \rangle_S \leq 0$ and is closed by the same argument. **The remaining reduction is the nine positive-delta families.** As a structural consistency check, {1,2,3,4} activates four conditions with delta sum $+3 > 0$ — the generic argument correctly cannot close the extremal set, and a red control keeps it that way.

4. The template for a family

Each family is closed by the same three moves, one level down:

1. **A second-level dot theorem** on the family's own cone: a weight and an anchored equispaced set chosen so that the anchored members contribute ≤ -1 exactly and the base inner product is ≤ 0 ; the collision conditions are again derived, and any positive-delta condition defines a **subfamily** with one fewer degree of freedom. Where the family's region is not a single cone (a and c-b incomparable, parity constraints), it is first **triangulated** into finitely many cones, checked onto point-by-point over a box (and, in Section 5, argued once each).
2. **Anchored closures** (Section 2.4) for every subfamily, with derived thresholds; a subfamily whose two parameters are independently unbounded gets a union of two closures, one per tail.

3. **Finite decisions** below the thresholds, one certificate per set. In the three families containing $\{1,2,3,4\}$ (the equality families $d = 2b$, $d = a+c$, $2d = 2c+b$), the extremal set appears exactly once, in a single cone or ray, where the finite decision SKIPS it as the definitional witness — equality, not a violation — and certifies every neighbour strictly below the target.

Two conditions of the family $2d = 2c+b$ are not closed by new work at all: their sets satisfy $d = 2a$ (respectively $d = a+b$) identically, families already closed — the record marks them DELEGATED. In total 2231 finite sets were decided across the nine families, zero undecided, and the only skipped set — ever — is $\{1,2,3,4\}$.

5. The nine families

The cone parametrizations below are bijections onto the stated regions; each is the standard prefix/chain substitution (new variables = successive differences, each ≥ 1), composed where noted with a parity substitution ($a = 2t$ etc.) or a triangulation on the one incomparable pair. Onto-ness is checked point-by-point over a box at every campaign run; the inverse maps are stated inline. Derived data (thresholds, floors, finite counts) below is read from the record.

5.1 Family $d = 2c$

Region $a < b < c$ free (prefix cone; $d = 2c$ derived). Dot on $S(c, 2\pi/3)$: anchored c, d at $-1/2$ each; $w = 2(1-\cos a)^2 + 2(1-\cos b)^2$; base $-2/5$. Four positive conditions.

Closures and finite parts (from the record):

- $b = 2a$: $N0 = 19$ (floor 12) OR $N0 = 19$ (floor 12) ; finite 50 enumerated / 50 certified - $c = 2a$: $N0 = 3$; finite 1 enumerated / 1 certified - $c = 2b$: $N0 = 3$; finite 1 enumerated / 1 certified - $c = a+b$: $N0 = 25$ (floor 12) ; finite 179 enumerated / 179 certified

5.2 Family $d = 2a$

Region $a < b < c < 2a$: chain with the substitution $a = (c-b) + (b-a) + r$, all differences ≥ 1 — inverse $r = 2a - c$. Dot on $S(a, 2\pi/3)$: anchored a, d at $-1/2$; $w = 2(1-\cos b)^2 + 2(1-\cos c)^2$; base $-2/5$. EVERY condition is negative: the family closes generically.

Closures and finite parts (from the record):

5.3 Family $d = b+c$

Prefix cone. Dot on $S(b+c, \pi)$: anchored d at -1 ; $w = 2(1-\cos a)^2$; base $-1/5$. One condition, negative (paired collision events on $2a = c$ and $2a = b$ cancel exactly): closes generically.

Closures and finite parts (from the record):

5.4 Family $d = a+b$

Region $c < a+b$: substitution $a = (c-b) + r$. Dot on $S(a+b, \pi)$: $w = 2(1-\cos c)^2$; base $-1/5$. One positive condition, $2d = 3c$ — the doubly constrained subfamily $\{3p+q, 3p+2q, 4p+2q, 6p+3q\}$ (inverse: $q = b-a$, $p = (2a-b)/3$). On $S(\gamma, \pi/3)$, $\gamma = d/3$, the four members are

CONSTANT at $-3/2$ — above the target; the closure instead evaluates on $S(\gamma, \pi/2)$ where c anchors at -1 , d at 0 , a lands at π (Lemma D) and b at $\pi/2$ (Lemma F).

Closures and finite parts (from the record):

- E: $2d = 3c$: $N_0 = 8$; finite 8 enumerated / 8 certified

5.5 Family $d = a+c$

Prefix cone; EQUALITY family. Dot on $S(a+c, \pi)$: $w = 2(1-\cos b)^2$; base $-1/5$. Two positive conditions: the arithmetic-progression subfamily $\{b-t, b, b+t, 2b\}$ ($2b = a+c$), closed by the reflection estimate — evaluation on $S(b, \pi)$ with companion $S(t, 0)$, so a and c land at π (Lemma D) and $d = 2b$ at $\cos 2\pi = +1$, giving $-2 + \pi^2/N^2$; and $3b = 2d$, closed on $S(b/2, \pi/2)$. The extremal set lives in the AP subfamily and is the skipped witness.

Closures and finite parts (from the record):

- AP: $2b = a+c$: $N_0 = 5$; finite 5 enumerated / 4 certified / skipped 1,2,3,4 - $3b = 2d$: $N_0 = 8$; finite 17 enumerated / 17 certified

5.6 Family $2d = 2c+a$

Parity substitution $a = 2t$; chain cone $(t, b-a, c-b)$. Dot on $S(d, \pi)$: $w = 2(1-\cos a)^2$; base $-1/5$. Four positive conditions — read from the engine after a hand-translation error was caught by the coverage gate — with subfamilies $2c = 3a$, $b = 2a$ (two-closure union), $c = 2a$ (triangulated on b vs $3a$ into two cones and the single reduced ray $\{2,3,4,5\}$), and $2c = 2b+a$ (union on $S(a/2, \pi/2)$, Lemma F).

Closures and finite parts (from the record):

- $2c = 3a$: $N_0 = 6$ (floor 6) ; finite 9 enumerated / 9 certified - $b = 2a$: $N_0 = 13$ (floor 6) OR $N_0 = 44$ (floor 24) ; finite 144 enumerated / 144 certified - $c = 2a$, $b < 3a/2$: $N_0 = 7$ (floor 6) ; finite 11 enumerated / 11 certified - $c = 2a$, $b = 3a$: finite only ; finite 1 enumerated / 1 certified - $c = 2a$, $b > 3a$: $N_0 = 7$ (floor 6) ; finite 11 enumerated / 11 certified - $2c = 2b+a$: $N_0 = 14$ OR $N_0 = 17$; finite 39 enumerated / 39 certified

5.7 Family $d = 2b$

EQUALITY family; a and $c-b$ incomparable, so triangulate: D_1 ($a < c-b$), D_2 ($a = c-b$, i.e. $c = a+b$), D_3 ($a > c-b$) — chain substitutions with inverses $r = 2b-c$ and $e = a - (c-b)$. Dots on $S(b, 2\pi/3)$ for D_1/D_3 (anchored b, d at $-1/2$; $w = 2(1-\cos a)^2 + 2(1-\cos c)^2$; base $-2/5$); ten positive conditions consolidating to SIX subfamilies ($U_1 b=2a$; U_2 the AP family again; $U_3 c=3b-2a$; $U_4 2c=3b+a$; $U_5 2c=3b-a$, a 1090-set two-tail box; $U_6 c=2a$). D_2 closes directly by an anchored closure with the extremal set skipped.

Closures and finite parts (from the record):

- U_1 : $b = 2a$ (cone 1): $N_0 = 7$ (floor 6) ; finite 11 enumerated / 11 certified - U_1 : $b = 2a$ (cone 2): $N_0 = 7$ (floor 6) ; finite 11 enumerated / 11 certified - U_1 : $b = 2a$ (cone 3): finite only ; finite 1

enumerated / 0 certified / skipped 1,2,3,4 - U2: $c = 2b - a$ (AP) (cone 1): $N_0 = 5$; finite 5
enumerated / 4 certified / skipped 1,2,3,4 - U3: $c = 3b - 2a$ (cone 1): $N_0 = 19$ (floor 12) ; finite 50
enumerated / 50 certified - U4: $2c = 3b + a$ (cone 1): $N_0 = 19$ (floor 12) ; finite 50 enumerated / 50
certified - U5: $2c = 3b - a$ (cone 1): $N_0 = 32$ (floor 12) OR $N_0 = 57$ (floor 24) ; finite 1090
enumerated / 1090 certified - U6: $c = 2a$ (cone 1): $N_0 = 19$ (floor 12) ; finite 101 enumerated / 101
certified - cone D2 (direct closure): $N_0 = 13$ (floor 6) ; finite 45 enumerated / 44 certified / skipped
1,2,3,4

5.8 Family $2d = 2c + b$

EQUALITY family; parity $b = 2\beta$; triangulate on a vs β (cones i, ii, iii). Dots on $S(c, \pi)$ for
i/iii (anchored c at -1 ; $w = 2(1 - \cos a)^2$; g covers a, b, d ; base $-1/5$); cone ii ($a = \beta$) closes by a
two-closure union with the extremal set skipped. Two conditions DELEGATE (their sets satisfy d
 $= 2a$ resp. $d = a + b$); the rest: $d = b + 2a, c = a + b$ (three cones incl. the extremal ray), $d = 2c - 2a$ (on
 $S(c - a, \pi/2)$, Lemma F), $c = 2a, 2c = 3a$ (on $S(a/2, \pi/3)$).

Closures and finite parts (from the record):

- $d = b + 2a$: $N_0 = 13$ (floor 12) ; finite 22 enumerated / 22 certified - $c = a + b$ (cone 1): $N_0 = 19$
(floor 6) ; finite 101 enumerated / 101 certified - $c = a + b$ (cone 2): finite only ; finite 1 enumerated
/ 0 certified / skipped 1,2,3,4 - $c = a + b$ (cone 3): $N_0 = 19$ (floor 6) ; finite 101 enumerated / 101
certified - $d = 2a$: DELEGATED to family $d = 2a$ - $d = a + b$: DELEGATED to family $d = a + b$ - $d =$
 $2c - 2a$: $N_0 = 6$; finite 1 enumerated / 1 certified - $c = 2a$: $N_0 = 19$ (floor 12) ; finite 50 enumerated /
50 certified - $2c = 3a$: $N_0 = 7$; finite 5 enumerated / 5 certified

5.9 Family $2d = 3c$

Parity $c = 2\gamma$; seven cones (b vs γ , then a vs $p = b - \gamma$, then $j = a - p$ vs p). Dots on
 $S(\gamma, \pi/3)$ with the LINEAR weight $(1 - \cos a) + (1 - \cos b)$ against $g = 1/10 + \cos a + \cos b$:
anchored c at $-1/2$, d at -1 ; base $-4/5$. Cone W1 ($b < \gamma$) closes generically. Subfamilies: $2b =$
 3γ and $4a = 3c$ close on the $\pi/6$ anchor (members at multiples 3, 4, 6 of s — cosines 0, $-1/2$,
 -1 , all rational); $a + b = c$ on $S(\gamma, \pi/3)$; $2a = c$ on $S(\gamma, \pi/2)$; $a + b = d$ delegates; four
reduced rays decided directly. Note: two condition vectors in cone W3c define the same
hyperplane (unequal keys, equal content); both map to the same subfamily and their deltas co-
activate additively.

Closures and finite parts (from the record):

- $2b = 3\gamma$ (cone 1): $N_0 = 3$; finite 1 enumerated / 1 certified - $2b = 3\gamma$ (cone 2): $N_0 = 3$;
finite 1 enumerated / 1 certified - $2b = 3\gamma$ (cone 3): $N_0 = 3$; finite 1 enumerated / 1 certified -
 $2b = 3\gamma$ (ray 1): finite only ; finite 1 enumerated / 1 certified - $2b = 3\gamma$ (ray 2): finite only ;
finite 1 enumerated / 1 certified - $a + b = c$: $N_0 = 7$ (floor 6) ; finite 11 enumerated / 11 certified - a
 $= \gamma$: $N_0 = 4$; finite 3 enumerated / 3 certified - $4a = 3c$: $N_0 = 3$; finite 1 enumerated / 1
certified - $a + b = d$: DELEGATED to family $d = a + b$ - $\{1,3,4,6\}$ ray: finite only ; finite 1 enumerated
/ 1 certified - $\{4,5,6,9\}$ ray: finite only ; finite 1 enumerated / 1 certified - $\{2,4,6,9\}$ ray: finite only ;

finite 1 enumerated / 1 certified - {6,7,8,12} ray: finite only ; finite 1 enumerated / 1 certified - cone
W2 (direct closure): N0 = 4 ; finite 3 enumerated / 3 certified

6. Assembly

Every A in N'_4 is covered: if d avoids all fourteen conditions, or activates only negative-delta ones, the generic case gives $\min f_A \leq -8/5$. Otherwise A lies in at least one of the nine families, each of which certifies $\min f_A \leq L(1,2,3,4)$ for all its members (strictly, except the skipped witness). Hence $\sup_A L(A) = L(1,2,3,4)$, attained by $\{1,2,3,4\}$; that is, $\lambda(4) = -L(1,2,3,4)$. $\square$

7. What a referee should check

1. Lemmas A–F above (elementary; proofs included).
2. The positivity/bounded-multiples principles of 2.2, and additivity of deltas.
3. The cone parametrizations of Section 5 (each a one-line change of variables; the campaign additionally checks each onto over a box, point-by-point).
4. That the engine's derived data is what this document quotes — regenerate it: the record is rewritten from scratch by `node tools/run-lambda4-campaign.js` and this document by `node tools/build-lambda4-writeup.js`; the independent audit `node tools/audit-lambda4.js` re-walks every clause with no shared code.
5. The finite certificates: each is a certified enclosure of a specific minimum from the calibrated instrument; the audit re-certifies every finite-clause set in its box fresh.

References

- [M] I. Mercer, *Finite searches, Chowla's cosine problem, and large Newman polynomials*, INTEGERS 19 (2019), #A4 (arXiv:1709.06612). The statement, Section-5 strategy, and Lemmas C–E are his; this work executes and completes the strategy.
- B. Bedert, *Polynomial bounds for the Chowla Cosine Problem*, arXiv:2509.05260 (asymptotic side).
- Jin–Milojević–Tomon–Zhang, arXiv:2509.03490 (asymptotic side).